FASTSPEECH 2 TEXT-TO-SPEECH FOR ENGLISH AND TELUGU

Technical Report

Kambhampati Advaith

SUMMARY

I trained two separate FastSpeech 2 models, one for English (Svarah) and one for Telugu (IndicTTS Telugu), starting from the ming024/FastSpeech2 code base and the pretrained universal HiFi-GAN vocoder. The English model was trained for 24,000 steps on 4.86 hours of multi-speaker, noisy, accented speech (91 speakers). The Telugu model was trained for 20,000 steps on 4.94 hours from two studio-recorded speakers. Both models produce intelligible speech with audible artefacts. Both overfit early: the best validation loss was reached at step 3,000 to 4,000 for English and at step 2,000 for Telugu, and the checkpoints used for synthesis (24,000 and 20,000) are later than that. The report describes the changes made to the reference code, the data preparation, the loss curves, the generated speech, and the duration, pitch and energy predicted by the variance adaptor for every test sentence.

LIST OF FIGURES

S. No	Figure No.	Figure Caption	Page No.
1	Figure 1	English training loss and validation loss	12
2	Figure 2	Telugu training and validation loss.	12
3	E1	predicted duration, F0 and energy.	19
4	E2a	predicted duration, F0 and energy.	20
5	E2b	predicted duration, F0 and energy.	21
6	E3	predicted duration, F0 and energy.	22
7	E4	predicted duration, F0 and energy.	23
8	T1	predicted duration, F0 and energy.	24
9	T2a	predicted duration, F0 and energy.	25
10	T2b	predicted duration, F0 and energy.	26
11	T3	predicted duration, F0 and energy.	27
12	T4	predicted duration, F0 and energy.	28

LIST OF TABLES

S. No	Table No.	Table Caption	Page No.
1	Table 1	Data used. Hours after alignment are computed from the raw clip durations in the selection manifest, so they include a little leading and trailing silence. A first English selection restricted to the top 40 speakers gave only 3.45 h of training audio, so it was repeated with all speakers.	9
2	Table 2	Training configuration. Parameters are listed in section 1.1. The English run was interrupted several times (session resets) and restarted from the step-12,000 checkpoint; the logs contain overlapping segments and the plots use the last entry for each step.	10
3	Table 3	Validation loss at selected steps (total = sum of mel, PostNet mel, pitch, energy and duration terms).	13
4	Table 4	Test sentences, front-end results and listening notes. E3 is the sentence as given in the assignment, including the stray space.	16
5	Table 5	Summary of the predictions. English values are for speaker spk000 (measured mean F0 131 Hz in the training data); Telugu values are for the speaker with code 1 (measured mean F0 188 Hz).	18

TABLE OF CONTENTS

S. No	Contents	Page No.
1	**Implementation approach**	**6**
1.1	Reference and components used	6
1.2	Modifications to the reference code	6
1.3	Language-specific adaptation	7
2	**Data, preprocessing and training configuration**	**9**
2.1	Dataset selection and split	9
2.2	Preprocessing	10
2.3	Training configuration	10
3	**Training and validation loss**	**12**
3.1	English	13
3.2	Telugu	14
4	Generated speech: observations	15
5	**Variance adaptor analysis**	**18**
5.1	General patterns	18
5.2	English sentences	19
5.3	Telugu sentences	24
6	**English compared with telugu**	**29**
7	**Reproducibility**	**31**
8	**References**	**32**

1. IMPLEMENTATION APPROACH

1.1 Reference and components used

The model follows Ren et al., FastSpeech 2: Fast and High-Quality End-to-End Text to Speech (ICLR 2021). The code is the PyTorch implementation at github.com/ming024/FastSpeech2, which I used as a starting point and modified. The following parts were used from the reference implementation without changing the architecture.

- **Encoder:** 4 feed-forward Transformer (FFT) blocks, hidden size 256, 2 attention heads, convolution filter size 1024 with kernels [9, 1].
- **Variance adaptor:** three predictors (duration in log domain, pitch, energy), each with 256 filters, kernel size 3 and dropout 0.5; a length regulator that expands phones to mel frames; pitch and energy are quantised into 256 linear bins and added back as embeddings. Pitch and energy are modelled per phone.
- **Mel decoder and PostNet:** 6 FFT blocks (hidden 256) followed by a convolutional PostNet. Loss: L1 on the mel spectrogram before and after the PostNet, plus mean squared error on log duration, pitch and energy.
- **Vocoder:** the pretrained universal HiFi-GAN generator shipped with the reference repository. It was not retrained or fine-tuned, so audio quality is limited by this mismatch as well as by the acoustic model.

The English model has 35,194,689 parameters (91 speakers). The Telugu model has the same architecture with a 2-speaker embedding table, which gives 35,171,905 parameters (derived from the English count; the smaller embedding table is the only difference).

1.2 Modifications to the reference code

File	Change and reason
audio/stft.py, audio/audio_processing.py	Keyword arguments for current librosa (mel filter bank, pad_center). The hard-coded .cuda() calls were removed so that feature extraction runs on CPU.
preprocessor/preprocessor.py	Only speaker directories are traversed, and files without an alignment are skipped (the original reused the variables of the previous utterance in that case).
model/modules.py	The energy embedding used the pitch control value (p_control). It now uses e_control. This only affects energy control at inference time.

utils/model.py, utils/tools.py	torch.load with weights_only=False for current PyTorch; vocoder paths made relative to the package; energy targets cast to float32 (training crashed with a Double/Float mismatch).
text/telugu.py (new), text/symbols.py	Telugu text front end (section 1.3). Telugu phones are added to the symbol table with the prefix "tel". An earlier prefix containing an underscore was cut off in the aligner output, which left empty phone sequences; renaming fixed it.
text/english_frontend.py (new)	Number, date and time normalisation, spelled-out AM/PM, and sentence-level G2P (section 1.3).
utils/config.py, utils/pipeline.py (new)	One config.yaml per language (preprocess, model and train sections); explicit train/validation split from a manifest, with a check that no utterance appears in both sets.
inference_core.py (new)	Synthesis of the test sentences; duration, pitch and energy plots; predictions.json with all predicted values; optional speed, pitch and energy control sweeps.
English/, Telugu/ scripts	Per-language preprocess.py, train.py and inference.py wrappers and config.yaml, as required by the assignment layout.

1.3 Language-specific adaptation

English

Text is normalized (numbers, years, dates, times; "AM" and "PM" are spelled as letters) and converted to ARPAbet phones with g2p_en, which uses CMUdict and a neural model for unseen words. The reference code looked words up one at a time in a lexicon. I run G2P on the whole sentence instead, because g2p_en chooses the pronunciation of heteronyms from the part of speech: in test sentence E2 this gives K AA1 N D AH0 K T for the noun in E2a and K AA0 N D AH1 K T for the verb in E2b. Forced alignment used Montreal Forced Aligner (MFA) with the pretrained english_us_arpa acoustic model and dictionary.

Telugu

The reference code has no Telugu support, so I wrote a front end. It applies Unicode NFC normalisation, removes joiners, and expands digits, ordinals (such as 12వ) and H:MM times into Telugu number words

(the number-word tables are mine and have not been checked by a native speaker). A rule-based akshara parser converts the script to phones: a consonant takes the inherent vowel a unless followed by a vowel sign or virama; conjuncts follow the virama; the anusvara becomes the nasal that matches the following stop (ng, ny, nn, n, m); the visarga becomes h; the clusters క్ష and జ్ఞ are expanded. Punctuation becomes a pause token. A pronunciation dictionary built with the same rules was used to train an MFA acoustic model from scratch on the 6 hours of Telugu data, because I did not use a pretrained Telugu aligner.

2. DATA, PREPROCESSING AND TRAINING CONFIGURATION

2.1 Dataset selection and split

	English	Telugu
Dataset	Svarah (AI4Bharat): 6,656 accented-English utterances, 9.6 h, test split only (Hugging Face release)	IndicTTS Telugu, Hugging Face copy SPRINGLab/IndicTTS_Telugu: 8,576 utterances, two speakers (gender codes 0 and 1)
Filtering	Clip length 1.5 to 15 s, at least 3 words, plain- text transcript (no tags or odd symbols): 4,752 of 6,656 utterances kept	Clip length 1 to 15 s, at least 90% Telugu letters in the transcript
Speakers	The release has no speaker-id column, so a speaker is identified by the combination of the speaker-metadata columns (115 keys found; Svarah documents 117 speakers). The 91 speakers with the most usable audio were used (7.2 h available).	About 3 h drawn from each of the two speakers, because neither speaker alone has the 6 h needed. The model is therefore two- speaker.
Sampling	6 h sampled at random (seed 13), split by utterance 5 h / 1 h, stratified by speaker	Same procedure (seed 13): 6 h total, split 5 h / 1 h by utterance, stratified by speaker
Selected	3,051 train (4.908 h) + 652 val (1.092 h)	2,164 train (4.998 h) + 440 val (1.004 h)
After alignment	3,678 of 3,703 files aligned (99.3%). Final: 3,026 train (4.86 h), 645 val (1.08 h)	2,581 of 2,604 files aligned (99.1%). Final: 2,143 train (4.94 h), 438 val (1.00 h)
Overlap	None: the split is explicit and checked in code. Validation shares speakers with training (utterance-disjoint, not speaker-disjoint).	Same

Table 1. Data used. Hours after alignment are computed from the raw clip durations in the selection manifest, so they include a little leading and trailing silence. A first English selection restricted to the top 40 speakers gave only 3.45 h of training audio, so it was repeated with all speakers.

2.2 Preprocessing

- Audio is resampled to 22,050 Hz mono and peak-normalised to 0.95. Each utterance gets a text file with the normalised transcript.
- MFA produces phone-level alignments (TextGrids). English used the pretrained model. Telugu used an acoustic model trained on the 6 h with the dictionary from section 1.3. Files without an alignment are dropped.
- **Features:** 80-bin mel spectrogram (FFT size 1024, window 1024, hop 256, 0 to 8,000 Hz, as required by the pretrained vocoder); durations from the alignment; phone-level F0 (pyworld) and energy (L2 norm of the STFT magnitude), both normalised with dataset-wide mean and standard deviation. For English the F0 statistics are mean 178.6 Hz, std 57.7 Hz, and the energy statistics are mean 47.8, std 38.0.
- **Measured speaker pitch (training data):** the English speaker used for synthesis (spk000) has mean F0 131 Hz (10th to 90th percentile 107 to 167 Hz). The two Telugu speakers measure 215 Hz (167 to 277 Hz) for code 0 and 188 Hz (129 to 249 Hz) for code 1. I did not verify the gender labels of the Telugu codes, so none is stated here.

2.3 Training configuration

	English	Telugu
Steps / epochs	24,000 steps, about 127 epochs (189 steps per epoch)	20,000 steps, about 149 epochs (134 steps per epoch)
Batch size, optimiser	16; Adam (betas 0.9 and 0.98, eps 1e-9), gradient clipping 1.0	Same
Learning rate schedule	Transformer schedule with 1,000 warm-ups steps, then multiplied by 0.3 at steps 12,000 and 18,000	Warm-up 1,000 steps, multiplied by 0.3 at steps 10,000 and 15,000
Validation / checkpoints	Validation every 1,000 steps; checkpoint every 4,000 steps	Same
Maximum length (max_seq_len)	1,000 frames (11.6 s). 155 of 3,026 training clips (5.1%) and 42 of 645 validation clips (6.5%) are longer than 11.6 s in raw duration; during training only the first 1,000 frames of such clips are used. Validation does not truncate.	1,500 frames (17.4 s): no clip is truncated

Hardware	Steps 0 to 12,000: Colab T4 (about 2.5 steps/s). Steps 12,000 to 24,000: Kaggle T4, resumed from the step-12,000 checkpoint (speed of the final run not recorded)	Kaggle (T4 x2 instance, one GPU used for this run)
Seeds	Data selection seed 13. Training seeds are not fixed in the reference code, so repeated runs will differ slightly.	Same

Table 2. Training configuration. Parameters are listed in section 1.1. The English run was interrupted several times (session resets) and restarted from the step-12,000 checkpoint; the logs contain overlapping segments and the plots use the last entry for each step.

3. TRAINING AND VALIDATION LOSS

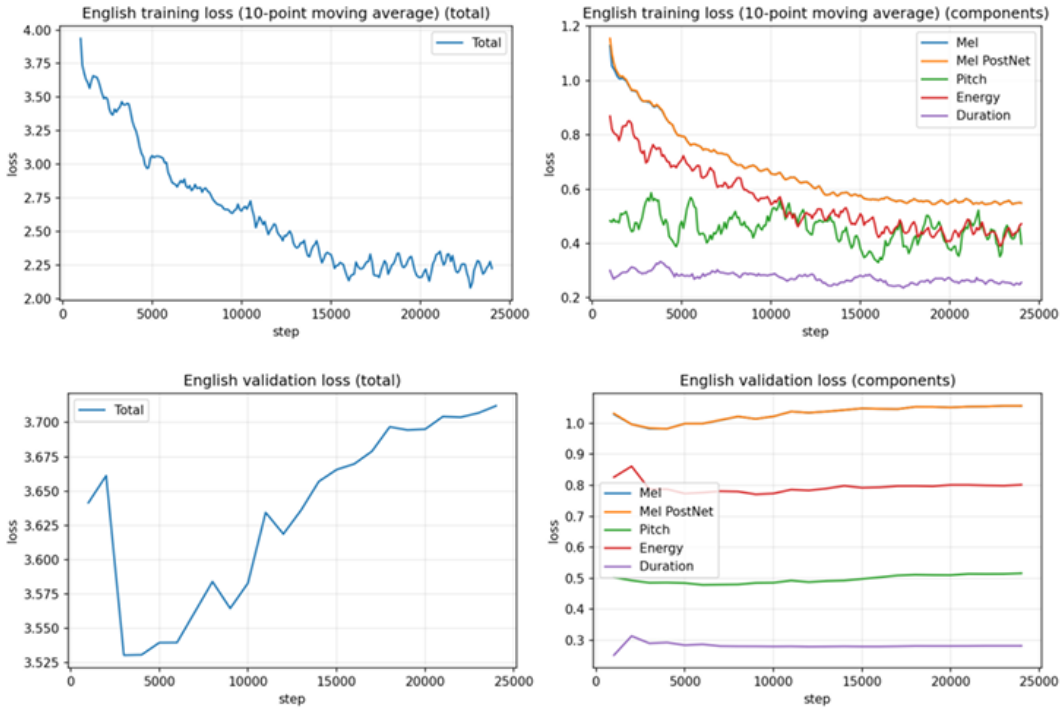


Figure 1. English training loss (10-point moving average of per-batch values, logged every 100 steps) and validation loss (every 1,000 steps).

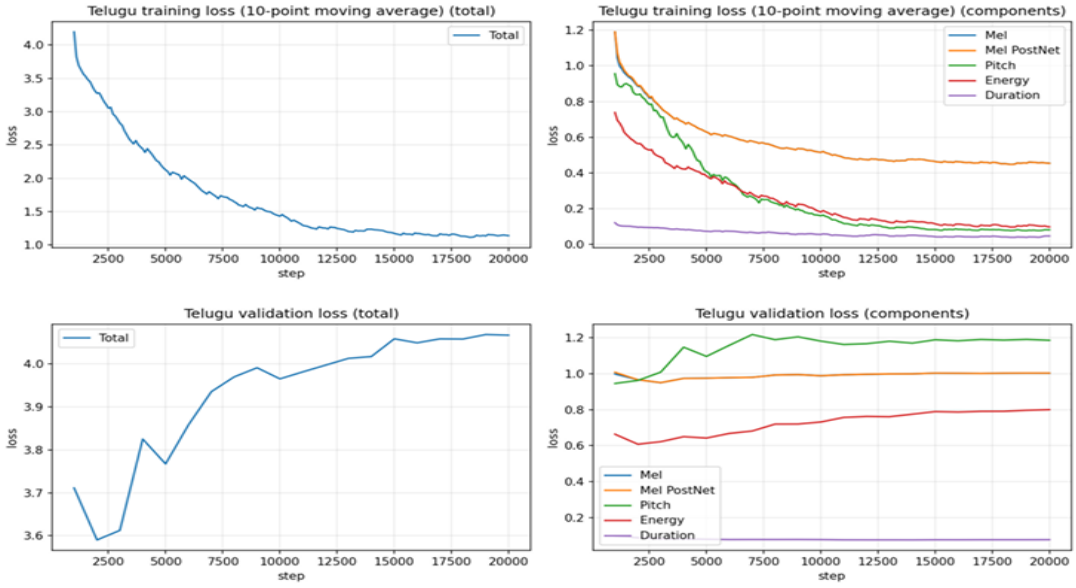


Figure 2. Telugu training and validation loss.

Step	English val total	English val mel	Telugu val total	Telugu val mel	Telugu val pitch
1000	3.642	1.029	3.711	0.998	0.945
2000	3.661	0.997	3.590	0.966	0.962
3000	3.530	0.983	3.613	0.949	1.008
4000	3.531	0.983	3.825	0.974	1.147
6000	3.540	0.999	3.858	0.978	1.157
8000	3.584	1.022	3.970	0.992	1.188
12000	3.619	1.034	3.998	0.996	1.166
16000	3.670	1.047	4.049	1.002	1.182
20000	3.695	1.051	4.067	1.002	1.185
24000	3.712	1.056	run ended	run ended	run ended

Table 3. Validation loss at selected steps (total = sum of mel, PostNet mel, pitch, energy and duration terms).

3.1 English

- **Convergence:** The training loss falls smoothly from 3.78 (mean over the first 2,000 steps) to 2.24 (last 6,000 steps) with no divergence or loss spikes. The resume at step 12,000 left no discontinuity (training loss 2.48 at step 12,000 and 2.27 at step 12,100).
- **Overfitting:** The validation total loss is lowest at 3.530 at step 3,000 (3.531 at step 4,000) and then rises steadily to 3.712 at step 24,000, 5.1% above the minimum. The mel loss follows the same pattern (0.983 at step 4,000, 1.056 at step 24,000), as does the pitch loss (0.479 at step 6,000 to 0.516). Energy rises slightly (0.776 to 0.802). The duration loss is flat at about 0.28 after step 2,000. The validation loss is measured with dropout off, so part of the gap to the training loss is not overfitting, but the rise while the training loss keeps falling is clear evidence of it.
- **Instability:** The per-batch training values are noisy (the pitch term varies by a factor of two between consecutive logs) because each logged value is a single batch. The moving average is smooth.
- **Consequence:** The checkpoint used for synthesis (24,000) is not the one with the lowest validation loss. I did not generate speech from earlier checkpoints, so I cannot say whether the audio would be better.

3.2 Telugu

- **Convergence:** The training loss falls from 4.26 (first 1,000 steps) to 1.15 (last 5,000 steps) and the mel loss from 1.21 to 0.46. There are no spikes.
- **Overfitting:** The validation total loss is lowest at 3.590 at step 2,000, then rises to 4.067 at step 20,000, which is 13.3% above the minimum. The validation mel loss stays near 1.0 while the training mel loss drops to 0.46, so the train/validation gap is much larger than for English. Possible reasons: only 2,143 training utterances (so about 149 passes over the data), and two speakers with clean, repetitive studio recordings that are easy to memorise.
- **Pitch and energy:** The validation pitch loss is 0.95 at step 1,000 and then stays between 1.0 and 1.2, which is about what a constant prediction would score if the normalised pitch has unit variance. The predicted average F0 still matches the speaker (section 6), so the model has learned the speaker level but not the sentence-level contour. The validation energy loss rises from 0.61 to 0.80. The duration loss is very low (0.078 against about 0.28 for English), which is consistent with the cleaner alignments of studio speech.
- **Consequence:** As with English, the final checkpoint (20,000) is far from the validation optimum. Earlier checkpoints were not compared.

4. GENERATED SPEECH: OBSERVATIONS

The listening notes below are mine, from listening to the generated files; there was no listening panel, no nativespeaker evaluation for Telugu and no objective score (such as an ASR round trip), so they should be read as informal judgements. English sentences were synthesised with speaker spk000 and Telugu sentences with the Telugu speaker of code 1. "Front-end" describes what the text front end produced, which is known exactly from the saved predictions.

ID	Sentence and front-end result	Listening notes	Speed and rhythm (predicted)
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings. The long word is converted to 28 plausible ARPAbet phones.	Clear. The long word came out clearly and was pronounced well.	6.48 s, 13.4 phones/s, no pauses.
E2a	The orchestra's conduct during the ceremony was praised by the audience. Noun reading: K AA1 N D AH0 K T.	The words were clear, but the sound was disturbed and not clean.	3.75 s, 13.6 phones/s, no pauses.
E2b	The researchers will conduct an experiment to evaluate the proposed method. Verb reading: K AA0 N D AH1 K T.	Everything was clear and correctly pronounced.	3.81 s, 14.7 phones/s (fastest English), no pauses.
E3	The rhyth m of the orchestra changed unexpectedly. The stray space is kept as given: G2P reads "rhyth" as R AY1 DH and "m" as the letter name EH1 M (standard "rhythm" is R IH1 DH AH0 M).	Heard as one word, with no audible break between the two parts. I did not check the vowel quality.	2.81 s, 13.2 phones/s, no pause token between the parts.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time. Expanded to "ten forty five A M", "twelfth November twenty twenty six"; "andaroo" gets AE0 N D AA1 R OW0.	"andaroo" was clear. The audio was somewhat disturbed. I did not judge the numbers separately.	5.80 s, 14.1 phones/s, one 70 ms pause at the comma.

ID	Sentence and front-end result	Listening notes	Speed and rhythm (predicted)
T1	న్యూ రోఫార్మ కోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు. The assignment writes the technical term as three words. The front end drops spaces, so the phones are the same as for the unsplit term (checked by eye, not by a native speaker).	Clear.	4.41 s, 15.0 phones/s, no pauses.
T2a	ాడుపై అతని పట్టు చాలా బలంగా ఉంద. పట్టు (grip) = p a tt tt u.	Not clear; the sound was disturbed.	2.09 s, 14.8 phones/s, no pauses.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చి ిం ది. పట్టు (silk) = p a tt tt u, the same phones as in T2a.	Clear.	2.44 s, 15.2 phones/s, no pauses.
T3	పులి అడ విలో నిశ్శ బ్దగం ా నడిచింది. Conjuncts and the geminate శ్శ handled as sh sh.	Clear, and fast.	1.88 s, 16.5 phones/s (fastest overall), no pauses.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి. Expanded to Telugu words: రెండు వేల ఇరవై ఆరు, పన్న ెండవ, పది నలభై ఐదు.	Clear. The date and the month sounded like English to the listener.	7.30 s, 16.4 phones/s, one 81 ms pause at the comma.

Table 4. Test sentences, front-end results and listening notes. E3 is the sentence as given in the assignment, including the stray space.

- **Where the English audio is disturbed:** E2a and E4 were reported as disturbed, E2b and E1 as clean. The predicted energy in E2a is at its lowest of the whole sentence between 1.16 and 1.29 s (about 5 to 6.5, against a sentence mean of 34), at the end of "conduct" and the start of "during". This is a candidate cause because very weak, stop-consonant-heavy stretches are hard for the vocoder, but I did not verify by listening that the artefact is located there.

- **Homographs:** For "conduct" the front end gives the correct, different phone sequences (noun and verb). For Telugu పట్టు the same phones are correct in both senses, so any difference between T2a and T2b can only come from context. In practice the predictions for the word are almost identical in the two sentences (27 frames each, mean F0 214 Hz against 211 Hz).
- **T4 - date and month:** The listener found T4 clear but heard the date and the month as English. The month word నవంబర్ is the English name "November" written in Telugu script (the front end gives it the phones n a v a m b a r), so an English-sounding result is expected for that word. The predicted phones for the date and the year are Telugu number words (పన్నెండవ, రెండు వేల ఇరవై ఆరు), so I cannot explain the English impression for those from the predictions alone.
- **The stray space in E3:** The model is given the sequence as written. Because "m" is read as a letter name, the result is an extra stressed syllable (F0 and energy both peak on EH1 at 0.48 s). The listener heard one word and no break, which fits the absence of a pause token.

5. VARIANCE ADAPTOR ANALYSIS

Each figure shows, for one test sentence, the predicted duration of every phone (in mel frames; one frame is 11.6 ms), and the predicted phone-level F0 and energy plotted over time. Grey bars and bands mark the pause token. Full-size versions are in the Results folder. F0 and energy are de-normalised with the training statistics. The table gives the main numbers.

ID	Phones	Length (s)	F0 mean (Hz)	F0 min to max (Hz)	Energy first 5 phones / rest	Longest phones (frames)
E1	87	6.48	128	120 to 147	80 / 26	AY0 12, Z 12
E2a	51	3.75	131	121 to 160	77 / 29	S 11, IY0 10
E2b	56	3.81	131	122 to 162	94 / 36	D 8, T 8, P 8
E3	37	2.81	128	121 to 141	73 / 42	AY1 16, IY0 11
E4	82	5.80	128	123 to 144	87 / 29	T 9, N 9
T1	66	4.41	197	128 to 273	91 / 54	uu 16, oo 13, ph 11
T2a	31	2.09	194	133 to 244	62 / 57	ai 13, aa 11, c 11
T2b	37	2.44	191	135 to 263	69 / 49	i 13, aa 11, c 11
T3	31	1.88	202	144 to 242	80 / 58	i 13, sh 11, sh 9
T4	120	7.30	177	96 to 245	76 / 45	ai 13, s 12, ii 11

Table 5. Summary of the predictions. English values are for speaker spk000 (measured mean F0 131 Hz in the training data); Telugu values are for the speaker with code 1 (measured mean F0 188 Hz).

5.1 General patterns

- **English duration is almost flat:** Nearly every phone is 4 to 9 frames (46 to 105 ms). The model has learned some phrase-final lengthening (the last phone is 11 to 12 frames in E1, E2a and E3) and a long stressed vowel in E3, but very little else.
- **English pitch is nearly monotone:** The predicted F0 of every sentence follows one shape: a peak of 141 to 162 Hz over the first content word, a plateau near 122 to 130 Hz, and a slow fall to about 120 Hz. The spread within a sentence is only 20 to 40 Hz (standard deviation 5 to 10 Hz), against a 60 Hz 10th-to-90thpercentile range for this speaker across the training data. The predicted mean (128 to 131 Hz) matches the speaker's measured mean (131 Hz).
- **English energy** starts high (first five phones about 80 against about 30 for the rest) and then stays low and fairly flat, with irregular local rises that do not simply follow lexical stress (in E2b the stressed vowels of “experiment" and "evaluate" have energy of only 16 and 18). A start-of-sentence peak this large is probably an artefact of the training data and the alignment boundaries, but I have not tested that.
- **Telugu is much more varied:** Durations range from 1 to 16 frames, with many 3-frame phones. F0 varies by 98 to 149 Hz within a sentence (standard deviation 26 to 34 Hz), compared with a 120 Hz

10th-to-90thpercentile range for this speaker. The predicted mean F0 (177 to 202 Hz) agrees with the speaker's measured 188 Hz. Energy swings between about 10 (consonant clusters) and about 130 (vowels).

- **Geminates and pauses:** In Telugu, double consonants are predicted as two long phones (for example sh 11 plus 9 frames in T3). Pauses at commas are short (70 ms in E4, 81 ms in T4), and in both languages the pitch does not reset or fall at the comma.
- **Odd predictions:** The Telugu nasal ny before c is predicted at one frame in all three places it occurs (T1 at 4.13 s, T2b at 1.28 s, T4 at 7.00 s), which removes it. A likely cause is that my G2P makes the anusvara a full nasal consonant while the speaker mostly nasalises the vowel, so the aligner gave it almost no time;

I have not tested this. In T4 the minimum F0 (96 Hz) falls on that one-frame phone, and the predicted energy goes slightly negative (-1.3 at 6.73 s) because the regression output is not clamped.

5.2 English sentences

The same figure layout is used for all ten sentences.

E1

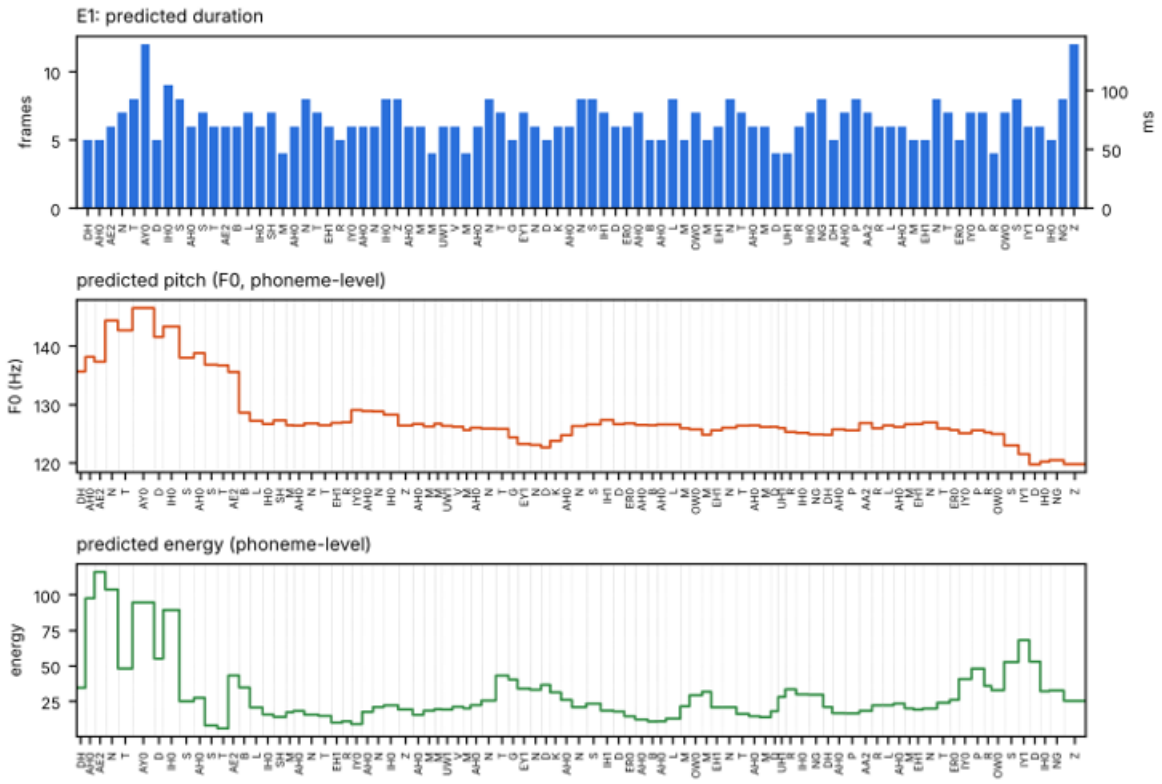


Figure. E1: predicted duration, F0 and energy.

- **Duration:** The median phone is 6 frames (70 ms). The only long phones are AY0 (12 frames, at 0.36 s) in the first syllables of "antidisestablishmentarianism" and the final Z of "proceedings" (12 frames, phrase final lengthening). Apart from the AY0, the 12-syllable word consists of the same 4 to 9 frame phones as the rest of the sentence, so it is spoken at an ordinary rate, which is consistent with its being heard clearly.
- **Pitch:** A peak of 147 Hz at 0.36 s on the start of the long word, then a drop to a plateau around 126 Hz and a final fall to 120 Hz. Standard deviation 5.5 Hz.
- **Energy:** Strongest at the start (mean 80 for the first five phones against 26 for the rest, peak 116 at 0.12s), then low and flat, with a rise to about 70 on the stressed syllable of "proceedings".

E2a

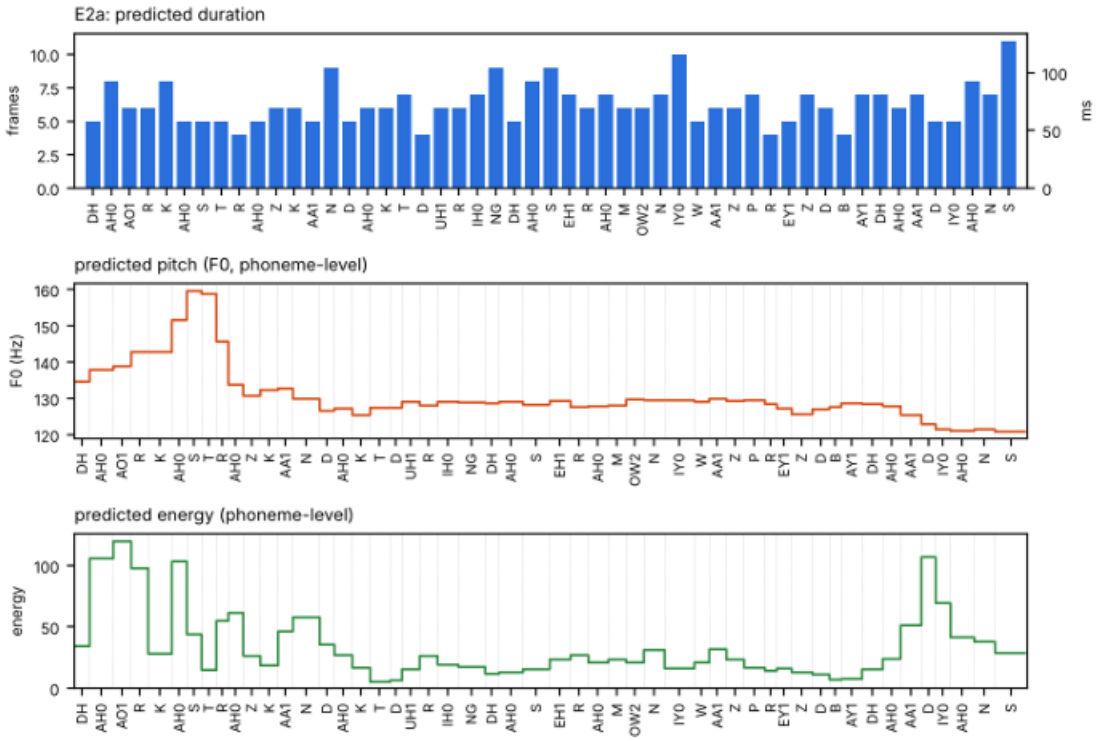


Figure. E2a: predicted duration, F0 and energy.

- **Duration:** Final lengthening on the last phone (S, 11 frames, 128 ms) and on the last vowel of "ceremony" (IY0, 10 frames at 2.32 s). The word "conduct" (0.73 to 1.24 s) has an ordinary profile (6, 5, 9, 5, 6, 6, 7 frames).
- **Pitch:** Peak of 160 Hz at 0.44 s in "orchestra's", then a plateau around 129 Hz and a fall to 121 Hz. The noun "conduct" (stress on the first syllable) sits on the plateau, mean 129 Hz and maximum 133 Hz, so there is no visible pitch accent.
- **Energy:** High at the start (77 against 29), a second peak of about 107 at the onset of "audience" (the D after AA1, at 3.33 s), and the lowest values of the sentence (5 to 6.5) at 1.16 to 1.29 s, at the end of

“conduct" and the start of "during". The listener found this sentence disturbed; this low-energy stretch is a plausible cause but unconfirmed.

E2b

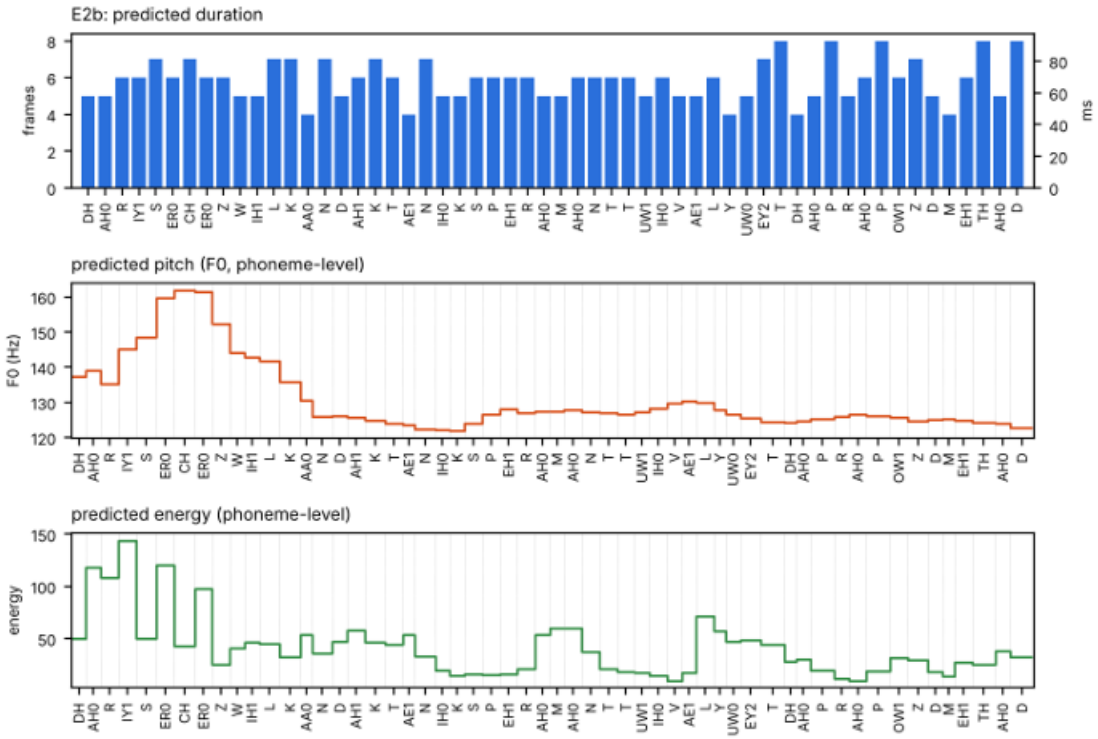


Figure. E2b: predicted duration, F0 and energy.

- **Duration:** The flattest and fastest English sentence (14.7 phones/s). The longest phones are only 8 frames (93 ms), including the final D, so there is little phrase-final lengthening.
- **Pitch:** Peak of 162 Hz at 0.41 s on "researchers", then 122 to 130 Hz. The verb "conduct" (stress on the second syllable) has a mean F0 of 128 Hz, almost the same as the noun in E2a (129 Hz). The noun/verb difference is therefore present in the phones but not in pitch. The mean energy over the word is higher in E2b (45 against 30), but I cannot tie that to stress.
- **Energy:** Strong start (94 against 36). There are local peaks up to about 70 (for example 71 on the L of "evaluate"), but the stressed vowels of "experiment" and "evaluate" themselves are low (16 and 18), so the predicted energy does not follow lexical stress.

E3

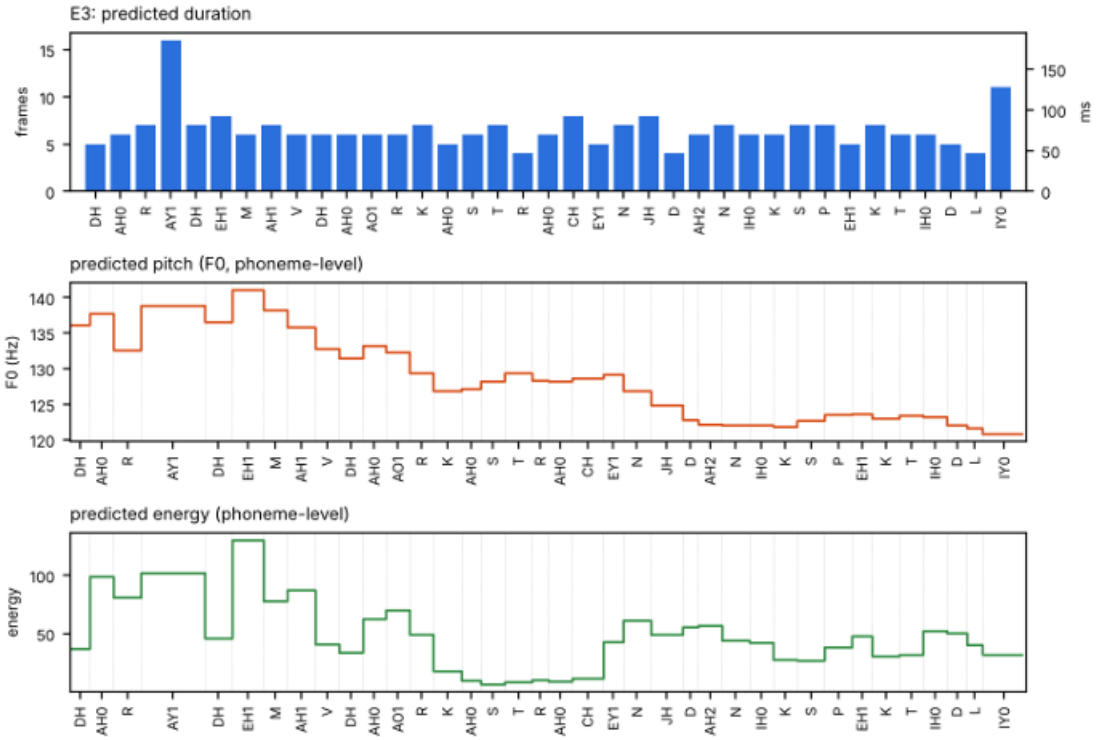


Figure. E3: predicted duration, F0 and energy.

- **Duration:** The longest phone of any English sentence is the stressed AY1 of "rhyth" (16 frames, 186 ms, at 0.21 s). The final IY0 of "unexpectedly" is 11 frames.
- **Pitch:** Peak of 141 Hz at 0.48 s, on the EH1 of the letter name "m", then a fall to 120 Hz. The model treats the letter-name syllable like a stressed syllable.
- **Energy:** The peak of the sentence (130) is on the same EH1 at 0.48 s. There is no pause between "rhyth" and "m", consistent with the listener hearing one word.

E4

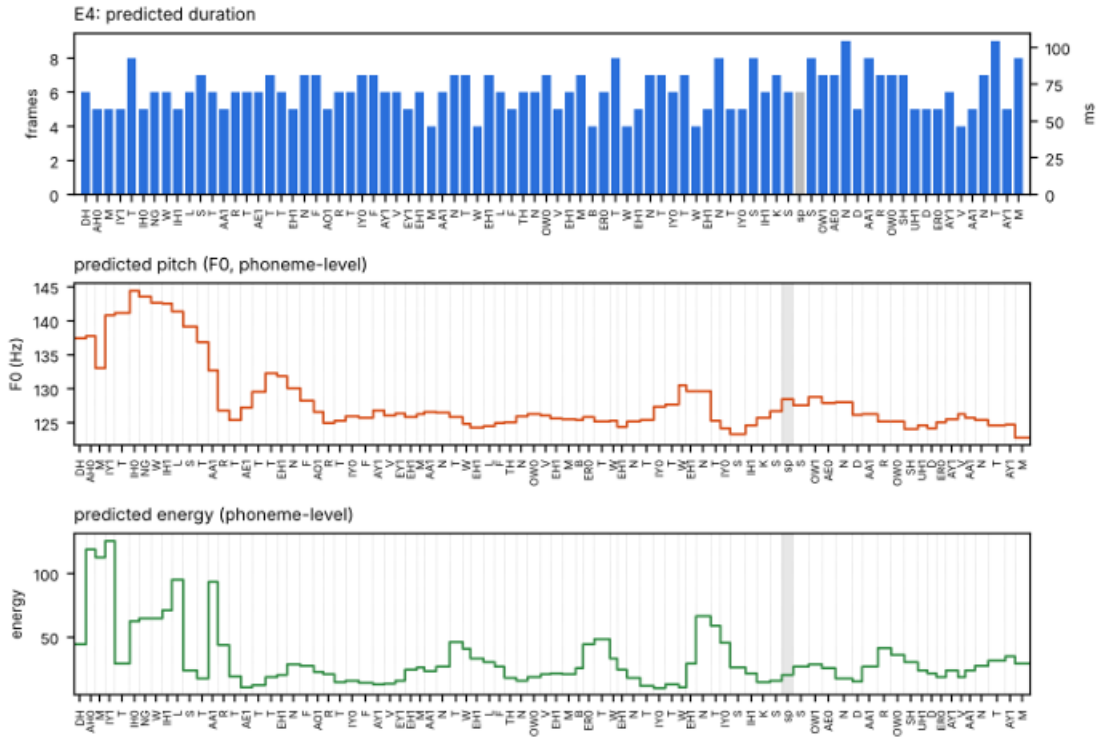


Figure. E4: predicted duration, F0 and energy.

- **Duration:** 82 phones in 5.80 s. The comma gives a 6-frame (70 ms) pause at 4.30 s, which is short. The longest phones are T of "time" (9 frames) and N of "andaroo" (9 frames).
- **Pitch:** The flattest sentence (standard deviation 5.3 Hz): a peak of 144 Hz at 0.34 s, then 123 to 132 Hz. At the comma the F0 is 127, 129 and 128 Hz before, at and after the pause, so there is no phrase-boundary movement.
- **Energy:** A strong start (87 against 29) and a shallow dip to about 20 at the pause. The numbers and the date are expanded as intended ("ten forty five A M", "twelfth November twenty twenty six").

5.3 Telugu sentences

T1

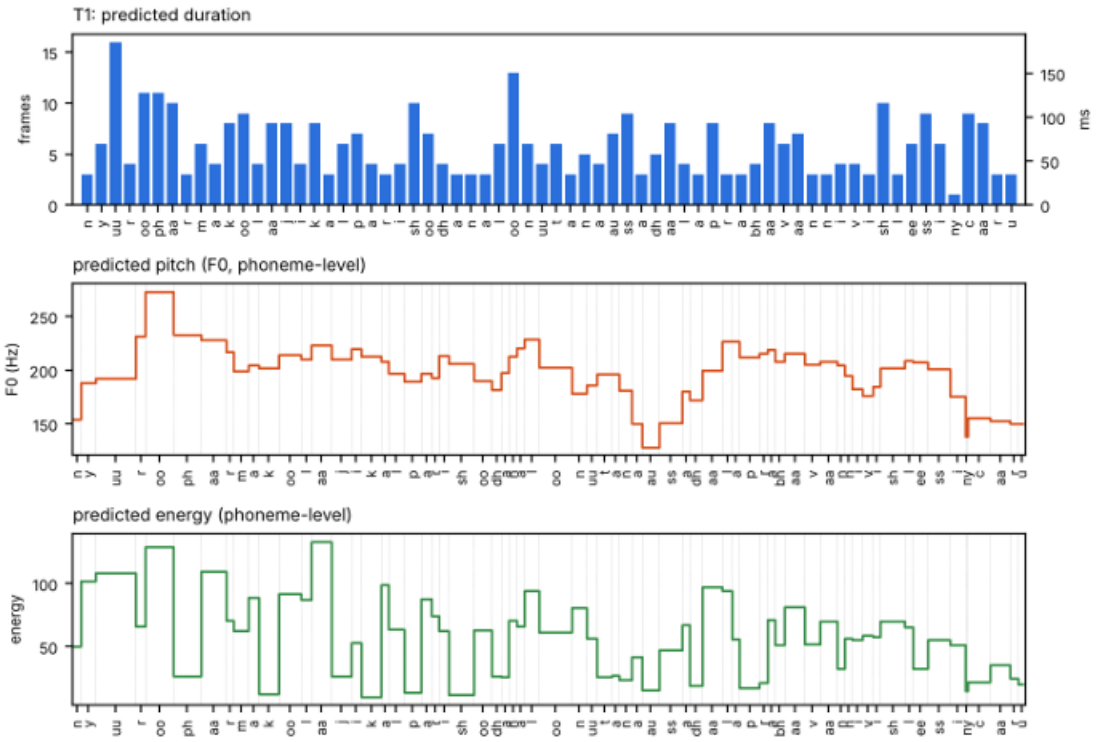


Figure. T1: predicted duration, F0 and energy.

- **Duration:** The longest phone is the uu of the first syllable (16 frames, 186 ms at 0.10 s). The word-final oo of పరిశోధనలో is lengthened (13 frames at 2.16 s) and ph is long (11 frames). The ny before the final c is 1 frame.
- **Pitch:** A peak of 273 Hz at 0.34 s on oo in the first word, a plateau around 200 Hz and a fall to about 149 Hz at the end. Standard deviation 25.7 Hz.
- **Energy:** Strong swings between about 10 and 133 (peak on aa at 1.10 s); the first five phones average 91 against 54 for the rest.

T2a

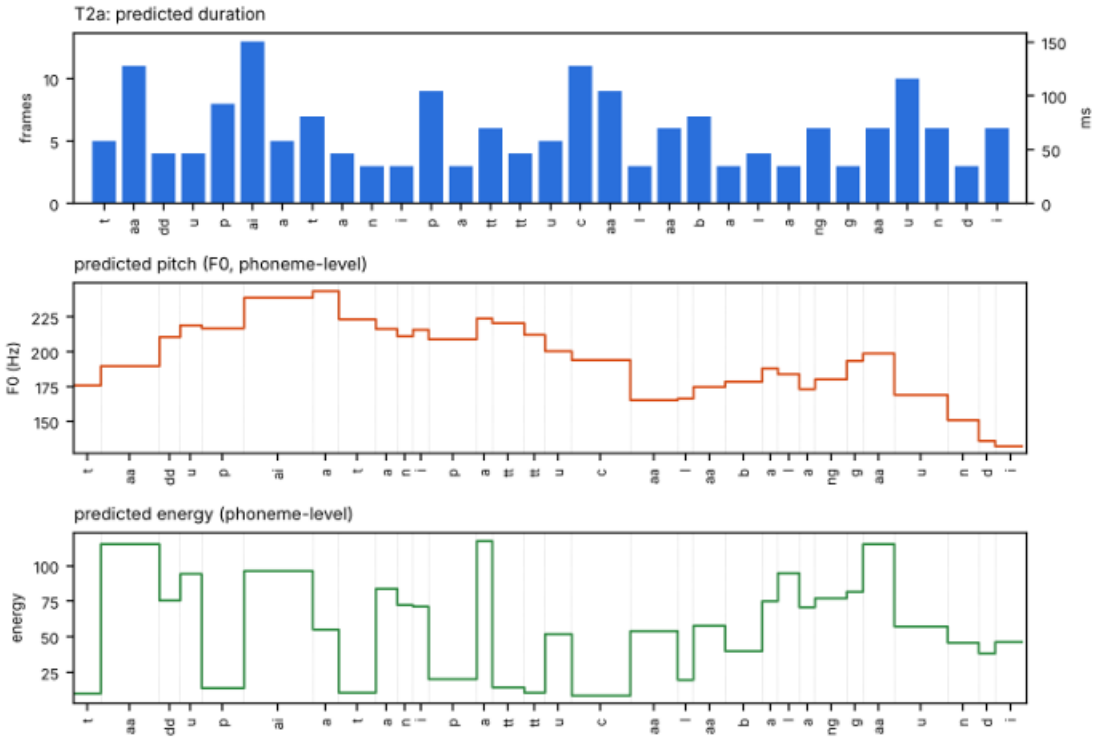


Figure. T2a: predicted duration, F0 and energy.

- **Duration:** A 13-frame ai at the end of తాడుపై (0.37 s), a long first vowel aa (11 frames) in తాడు, and a long c (11 frames) in చాలా. పట్టు is p a tt tt u with 9, 3, 6, 4, 5 frames (27 frames, 313 ms), the geminate tt tt being 6 plus 4 frames.
- **Pitch:** Peak of 244 Hz at 0.52 s and a steady decline to about 158 Hz at the end. The mean F0 over పట్టు is 214 Hz.
- **Energy:** The flattest Telugu energy profile at the start (62 for the first five phones against 57 for the rest), with a peak of 118 at 0.88 s. The listener found this sentence unclear and disturbed. I found no single cause in the predictions.

T2b

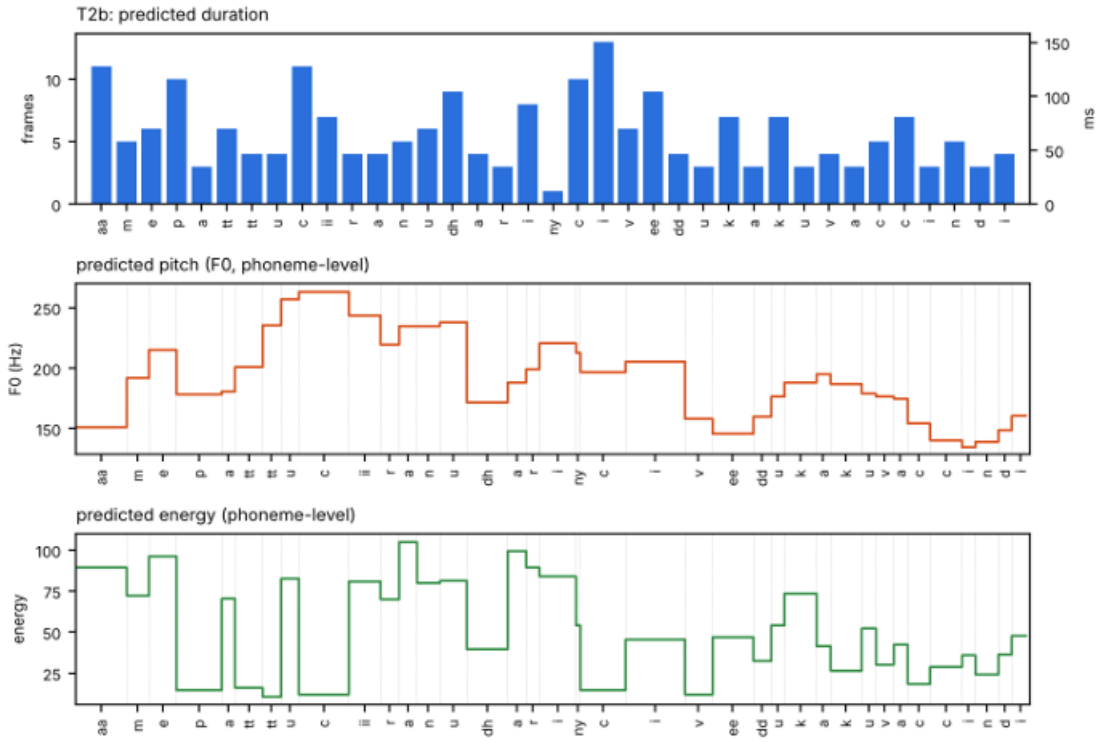


Figure. T2b: predicted duration, F0 and energy.

- **Duration:** Word-final i of ధరించి is 13 frames (1.40 s); c at the junction of పట్టుచీరను is 11 frames; the geminate c c in వచ్చింది is 5 plus 7 frames. The ny at 1.28 s is 1 frame. పట్టు gets 10, 3, 6, 4, 4 frames. The same shape as in T2a.
- **Pitch:** The largest pitch movement of the Telugu set (standard deviation 34 Hz): a peak of 263 Hz at 0.57s and a fall to 145 Hz at the end. Mean F0 over పట్టు is 211 Hz, almost the same as in T2a (214 Hz).
- **Energy:** Peak of 105 at 0.82 s; strongest in the first half of the sentence and lower toward the end.

T3

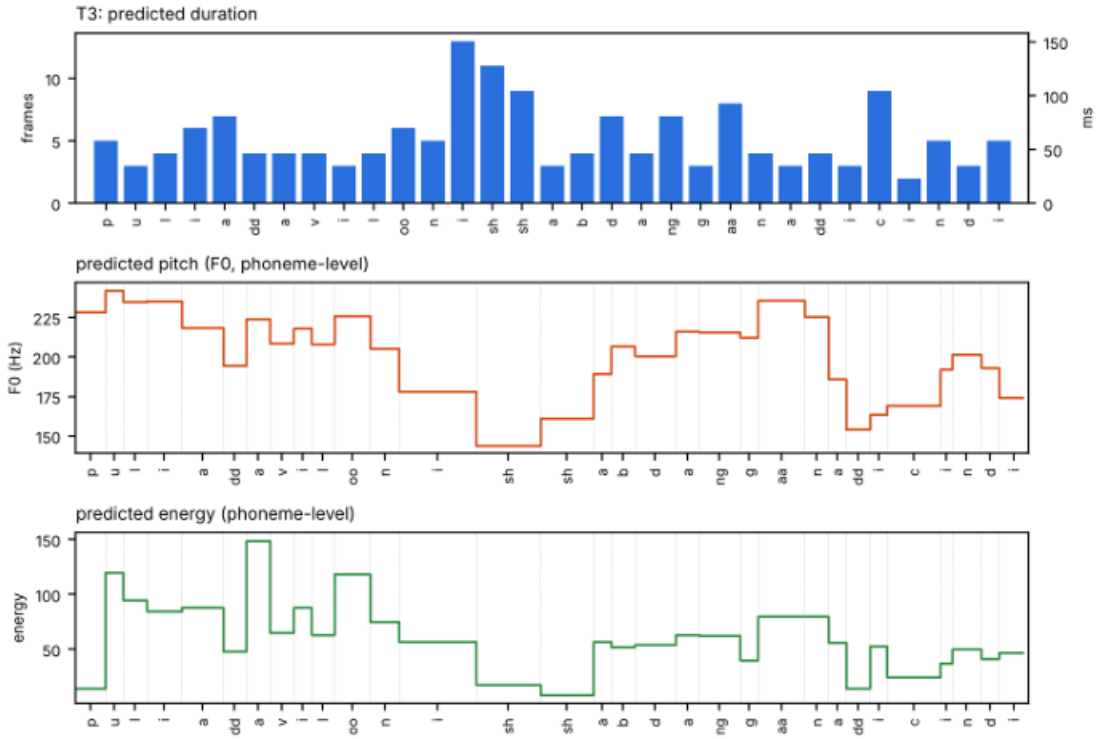


Figure. T3: predicted duration, F0 and energy.

- **Duration:** The fastest sentence overall (16.5 phones/s, median phone 4 frames). The syllable నిశ్శ takes i 13 frames and the geminate sh sh 11 plus 9 frames (232 ms for the double consonant).
- **Pitch:** It starts at 242 Hz on the first phone, drops to 144 Hz at 0.79 s (around the sh geminate), rises again to 236 Hz at 1.35 s in దంగా and ends at 175 Hz on the last phone (186 Hz averaged over the last five). This is the only Telugu sentence with a clear second pitch peak.
- **Energy:** Peak of 149 at 0.34 s; a very low stretch (about 8 to 18) over the geminate, then a recovery.

T4

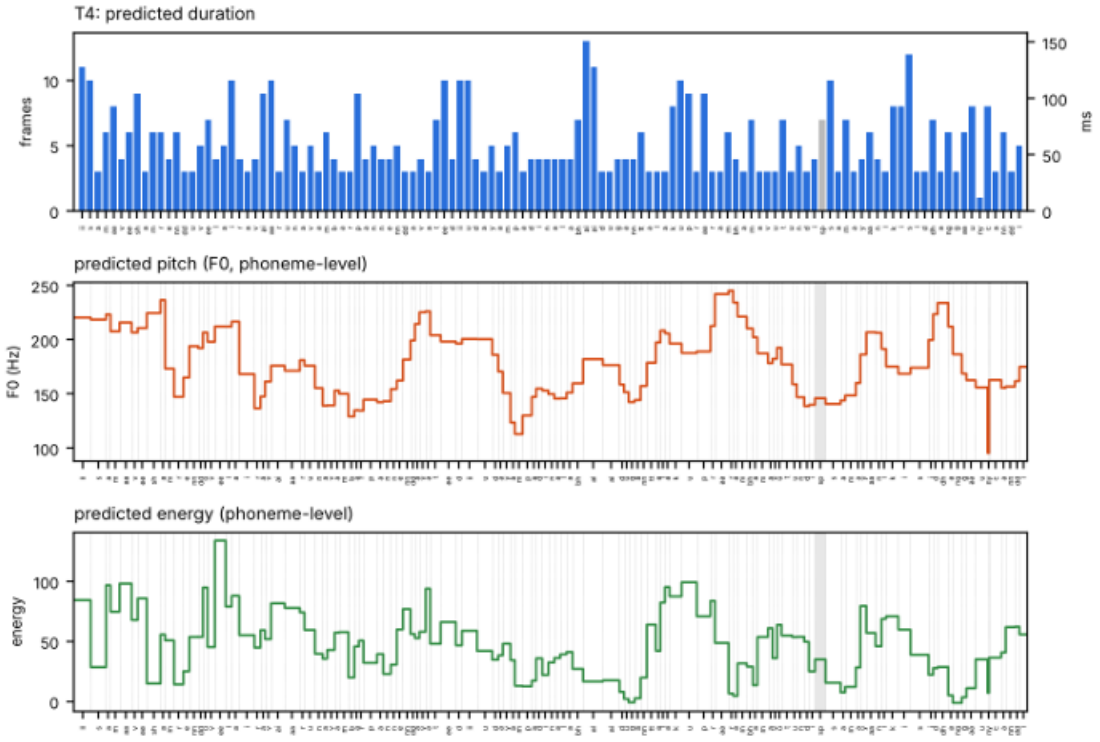


Figure. T4: predicted duration, F0 and energy.

- **Duration:** 120 phones in 7.30 s. The comma pause is 7 frames (81 ms) at 5.68 s. The word-final ai of నలభై is the longest phone (13 frames at 3.90 s). The ny in ఉంచండి is 1 frame at 7.00 s.
- **Pitch:** A peak of 245 Hz at 5.02 s (ప్రారంభం), a start-of-sentence level of 217 Hz and a fall to 162 Hz at the end. The F0 is 140, 146 and 141 Hz before, at and after the comma. The 96 Hz minimum at 7.00 s coincides with the one-frame ny and is probably an artefact.
- **Energy:** Peak of 134 at 1.08 s; the predicted value dips below zero (-1.3 at 6.73 s, near the end of సిద్ధంగా), which is not physically meaningful. The date and time were expanded to Telugu number words as intended.

6. ENGLISH COMPARED WITH TELUGU

	English (Svarah)	Telugu (IndicTTS)
Data	Noisy, accented, many speakers (91) with little audio each, speaker identities inferred from metadata	Studio quality, two speakers, about 3 h each, repetitive sentence style
Alignment	99.3% of files aligned with a pretrained model	99.1% aligned with a model trained on 6 h; duration loss 0.078 against 0.28
Validation loss	Min 3.530 (step 3,000), final 3.712 (+5.1%)	Min 3.590 (step 2,000), final 4.067 (+13.3%)
Pitch	Mean F0 matches the speaker (128 to 131 Hz against 131 Hz); movement within a sentence only 20 to 40 Hz	Mean matches (177 to 202 Hz against 188 Hz); movement 98 to 149 Hz; validation pitch loss at or above 1.0
Duration	Nearly flat (4 to 9 frames), small final lengthening	1 to 16 frames, geminates and word-final lengthening visible
Listening	E1 and E2b clear; E2a and E4 disturbed	T1, T2b, T3 and T4 clear (T3 fast; in T4 the date and month sounded English); T2a unclear
Front end	Mature tools (CMUdict, g2p_en); the stray space in E3 causes a letter-name reading	Own rule-based G2P and number expansion, not validated by a native speaker; the nasal before c is lost

Limitations and failures:

- Both models overfit early and the final checkpoints were used. I did not compare earlier checkpoints, so I cannot say how much audio quality depends on this.
- The English training ignores the part of each clip beyond 11.6 s (about 5% of clips), because max_seq_len was left at 1,000.
- Two sentences (E2a, T2a) were reported as disturbed or unclear and one (E4) as somewhat disturbed. I did not locate the artefacts by listening, and the energy and duration explanations above are hypotheses.
- Energy and F0 are regressed without constraints, so impossible values (negative energy, a 96 Hz dip on a one-frame phone) occur.
- Pauses are too short, and pitch does not reset at commas.
- The vocoder is the universal pretrained one. It was not fine-tuned on either dataset.

- The evaluation is informal: one listener (the author), no native-speaker check for Telugu, no objective metrics.

Possible improvements:

Use early stopping or choose a checkpoint from the validation minimum, and compare it by listening; add regularisation or data augmentation; raise max_seq_len or drop long clips for English; finetune the vocoder on the target data; normalise pitch and energy per speaker; constrain energy to non-negative values; model the anusvara as nasalisation of the vowel; add a better pause model driven by punctuation; and evaluate with an ASR round trip and a small listening test with native speakers.

7. REPRODUCIBILITY

Software: Python 3.13 (Colab and Kaggle runtimes). PyTorch as preinstalled on the runtime (the exact version was not recorded; run pip freeze to record it). Other packages are listed in requirements.txt: numpy, scipy, librosa (0.10 or later), pyworld, tgt, g2p_en, inflect, unidecode, pypinyin, nltk, scikit-learn, matplotlib, pyyaml, tensorboard, datasets (below 4), jiwer and transformers (the last two only for the optional ASR check). Montreal Forced Aligner was installed from conda-forge (version not recorded).

Configuration: English/config.yaml and Telugu/config.yaml hold the preprocessing, model and training settings used. The pretrained HiFi-GAN generator (generator_universal.pth.tar) comes from the reference repository.

Commands, run from the project root (details in README.md):

- *bash scripts/setup_env.sh*
- *python English/preprocess.py select (set HF_TOKEN first); bash scripts/run_mfa.sh English; python English/preprocess.py features*
- *python Telugu/preprocess.py select; bash scripts/run_mfa.sh Telugu; python Telugu/preprocess.py features*
- *python English/train.py and python Telugu/train.py (resume with --restore_step N)*
- *python scripts/plot_losses.py English and Telugu*
- *python English/inference.py --restore_step 24000 --controls and python Telugu/inference.py --restore_step 20000 –controls*

Checkpoints: The synthesis checkpoints are English step 24,000 and Telugu step 20,000, exported as weightsonly fp16 files (scripts/export_checkpoint.py) so that they fit GitHub's file-size limit. The repository URL and the checkpoint download link are given at the top of README.md.

Outputs: Results/English and Results/Telugu contain the loss plots, the generated audio, the variance-adaptor plots (duration, pitch, energy), spectrograms, predictions.json and the control experiments (speed, pitch and energy sweeps).

8. REFERENCES

1. Ren, Y., Hu, C., Tan, X., Qin, T., Zhao, S., Zhao, Z., and Liu, T. (2021). FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. International Conference on Learning Representations (ICLR 2021). Link: https://arxiv.org/abs/2006.04558

2. Ming024. FastSpeech 2 - PyTorch Implementation. GitHub Repository. Link: https://github.com/ming024/FastSpeech2

3. Kong, J., Kim, J., and Bae, J. (2020). HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. Advances in Neural Information Processing Systems (NeurIPS 2020). Link: https://arxiv.org/abs/2010.05646

4. McAuliffe, M., Socolof, M., Mihuc, S., Wagner, M., and Sonderegger, M. (2017). Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi. Proceedings of Interspeech 2017, pp. 498–502. Link: https://www.isca-archive.org/interspeech_2017/mcauliffe17_interspeech.html

5. AI4Bharat. (2023). Svarah: Evaluating English ASR Systems on Indian Accents. Indian English Speech Dataset.
Links:

- https://github.com/AI4Bharat/Svarah
- https://huggingface.co/datasets/ai4bharat/Svarah

6. Indian Institute of Technology Madras. IIT Madras TTS Database. Speech Corpus Resource. Link: https://indicnlp.ai4bharat.org/resources/